Poisson Distribution

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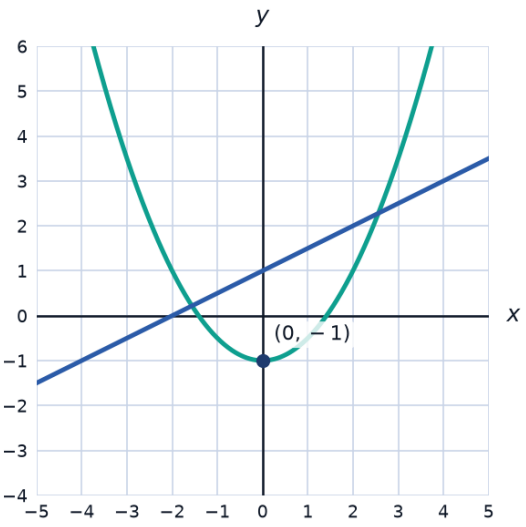
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

2 **[Foundational] • Target: Conditions for a Poisson distribution**

Explain why the number of texts a person receives in a randomly chosen hour might reasonably be modeled by a Poisson distribution, but the number of texts received right around a pre-arranged deadline might not.



3 **[Foundational] • Target: Computing Poisson probabilities**

A shop receives an average of 3 customers per minute, modeled by $X \sim \text{Po}(3)$. Find $P(X = 5)$, to three decimal places.

4 **[Foundational] • Target: Computing Poisson probabilities**

Using the same shop, find $P(X = 1)$, to three decimal places.

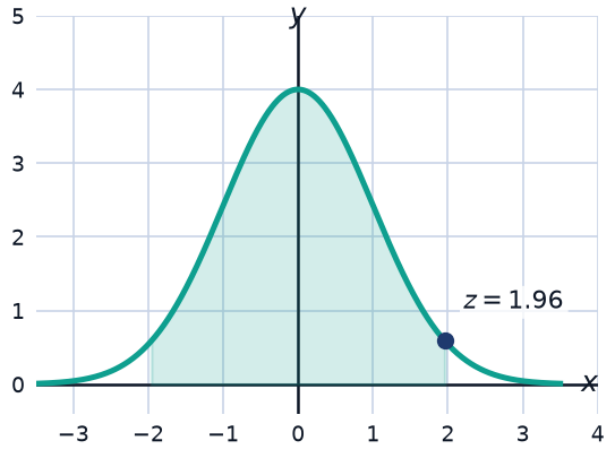
5 **[Foundational] • Target: Computing Poisson probabilities**

Using the same shop, find $P(X \leq 2)$.

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6 [Foundational] • Target: Mean and variance of a Poisson distribution

State the mean and variance of $X \sim \text{Po}(\lambda)$, and find the standard deviation when $\lambda = 9$.



7 [Foundational] • Target: Mean and variance of a Poisson distribution

A random variable $X \sim \text{Po}(7)$. State $E(X)$, $\text{Var}(X)$, and the standard deviation, to three decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums of independent Poisson variables

Two independent Poisson processes have rates $\lambda_1 = 4$ and $\lambda_2 = 6$ per hour. State the distribution of the total number of events, and find its mean and variance.

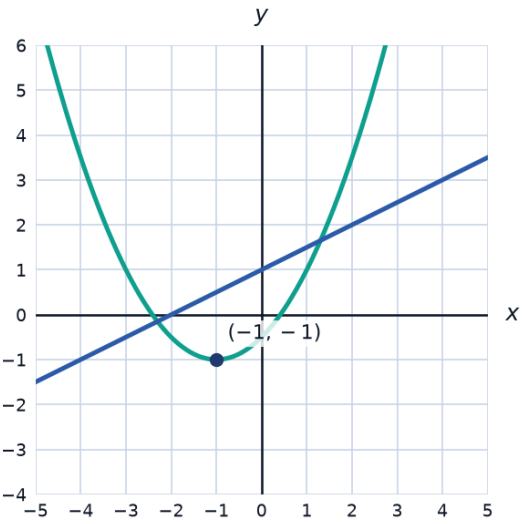
9 [Intermediate] • Target: Sums of independent Poisson variables

Three independent Poisson processes have rates $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 5$ per hour. Find the mean and variance of the total number of events across all three processes in one hour.

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10 [Intermediate] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the value of λ to use.



11 [Intermediate] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.0025. Using the Poisson approximation, find the probability that exactly 2 items are defective, to three decimal places.

12 [Intermediate] • Target: Poisson as an approximation to the binomial

A survey mails 5000 questionnaires, each independently containing a printing error with probability 0.0006. Using the Poisson approximation, find the probability that no questionnaires contain an error, to four decimal places.

13 [Intermediate] • Target: Using cumulative Poisson probabilities

A call center receives calls at $\lambda = 6$ per 10-minute interval. Find the probability of receiving more than 3 calls, given $P(X = 0) \approx 0.00248$, $P(X = 1) \approx 0.01487$, $P(X = 2) \approx 0.04462$, $P(X = 3) \approx 0.08924$.

14 [Intermediate] • Target: Using cumulative Poisson probabilities

A website receives visitors at $\lambda = 4$ per minute. Find $P(1 \le X \le 3)$, given $P(X = 0) \approx 0.0183$, $P(X = 1) \approx 0.0733$, $P(X = 2) \approx 0.1465$, $P(X = 3) \approx 0.1954$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

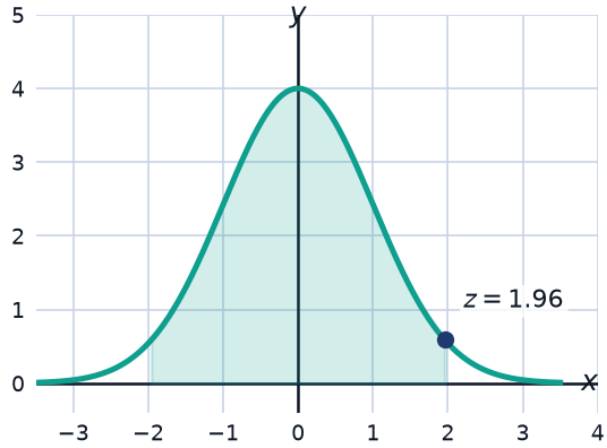
15 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 13.01 Continuous Random Variables

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.0498$. Find λ, to the nearest whole number.

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16 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 14.01 Sampling & Estimation

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 2. Find λ, and then find $P(X = \lambda)$, to three decimal places.



17 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 15.01 Hypothesis Testing

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 1) = 0.1494$, where λ is a whole number less than 5. Using $P(X = 1) = \lambda e^{-\lambda}$, determine λ by trial.

18 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 2.01 Exponentials & Logarithms

Machine faults occur at an average rate of $\lambda = 2$ per day. Find the probability that exactly 2 faults occur today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.1353$ and $P(X = 2) \approx 0.2707$.

19 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 3.01 Advanced Trigonometry

A radioactive source emits particles at a rate of $\lambda = 1.2$ per second, modeled by a Poisson distribution. Find $P(X = 2)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

20 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 4.01 Differentiation

This question has four parts. Faulty caps occur at random in bottling cases with an average rate of $\lambda = 1.5$ per case. (a) Find $P(X = 0)$ and $P(X = 1)$ for a randomly selected case, to four decimal places. (b) Find the probability that a case has at least 2 faulty caps. (c) The plant inspects 40 cases per day. Using the fact that a sum of independent Poisson variables is itself Poisson, state the distribution of the total number of faulty caps found across all 40 cases in a day, and find its mean. (d) Since this total has a large λ, it can be approximated by $N(\lambda, \lambda)$. Using a continuity correction and $\Phi(1.36) \approx 0.9131$, estimate the probability that more than 70 faulty caps are found in a day.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Poisson Probability Formula & Conditions

Which of the following is NOT an essential assumption for modeling a random variable with a Poisson distribution $\text{Po}(\lambda)$?

a. Events occur at a variable rate over time

b. Events occur singly and independently

c. The average rate of occurrence λ is constant

d. Two events cannot occur at the exact same instant

22 [Foundational] • Target: Poisson Probability Formula & Conditions

The random variable $X \sim \text{Po}(3.2)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.2087

b. 0.2226

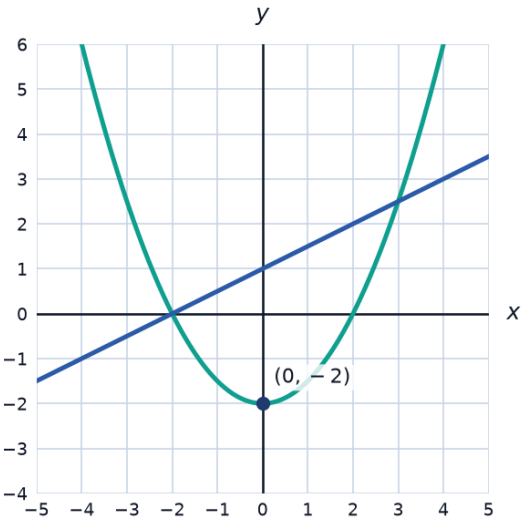
c. 0.1782

d. 0.1450

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23 [Foundational] • Target: Poisson Probability Formula & Conditions

If $X \sim \text{Po}(\lambda)$ and $P(X = 0) = 0.05$, find the value of λ.



a. $\lambda = -\ln(0.05) \approx 2.996$

b. $\lambda = 0.05$

c. $\lambda = e^{0.05} \approx 1.051$

d. $\lambda = 20$

24 [Foundational] • Target: Mean, Variance & Linear Combinations

What is the theoretical relationship between the mean and variance of a Poisson distribution $X \sim \text{Po}(\lambda)$?

a. $\text{Mean} = \text{Variance} = \lambda$

b. $\text{Mean} = \lambda, \text{Variance} = \lambda^2$

c. $\text{Mean} = \lambda, \text{Variance} = \sqrt{\lambda}$

d. $\text{Mean} = \lambda, \text{Variance} = \lambda(1 - p)$

25 [Foundational] • Target: Mean, Variance & Linear Combinations

Independent random variables $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(3)$ represent customer arrivals. What distribution does $X + Y$ follow?

a. $\text{Po}(5)$

b. $\text{Po}(6)$

c. $\text{Po}(2.5)$

d. $N(5, 5)$

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26 [Foundational] • Target: Mean, Variance & Linear Combinations

Faults on a fiber optic cable occur at an average rate of 0.4 per kilometer. What is the probability of finding exactly 1 fault in a 5-kilometer section?

a. 0.2707

b. 0.4000

c. 0.1353

d. 0.3250

27 [Foundational] • Target: Poisson Approximation to the Binomial

Under which conditions can a binomial distribution $B(n, p)$ be well approximated by a Poisson distribution?

a. $n > 50$ and $np < 5$

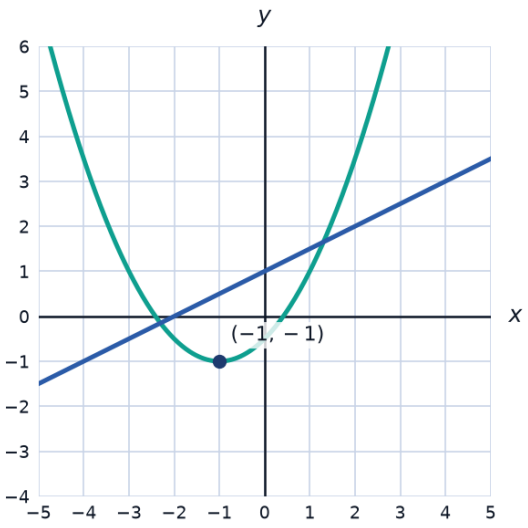
b. $n > 30$ and $p = 0.5$

c. $np > 5$ and $n(1 - p) > 5$

d. $n < 20$ and $p > 0.1$

28 [Foundational] • Target: Poisson Approximation to the Binomial

A book of 400 pages contains 200 typographical errors distributed at random. Use a Poisson approximation to find the probability that a randomly chosen page contains no errors.



a. $e^{-0.5} \approx 0.6065$

b. 0.5000

c. 0.3679

d. 0.7500

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Poisson Approximation to the Binomial

A factory produces screws, of which 1% are defective. In a random sample of 250 screws, what is the Poisson parameter λ to model defective screws?

a. $\lambda = 2.5$

b. $\lambda = 0.01$

c. $\lambda = 25$

d. $\lambda = 5$

30 [Intermediate] • Target: Normal Approximation to the Poisson Distribution

Under what condition can a Poisson distribution $X \sim \text{Po}(\lambda)$ be approximated by a normal distribution?

a. $\lambda > 15$

b. $\lambda < 5$

c. λ is an integer

d. $\lambda > 100$ only

31 [Intermediate] • Target: Normal Approximation to the Poisson Distribution

If $X \sim \text{Po}(36)$, which normal distribution provides the best approximation?

a. $N(36, 36)$ with mean 36 and variance 36

b. $N(36, 6)$

c. $N(0, 1)$

d. $N(36, 18)$

32 [Intermediate] • Target: Normal Approximation to the Poisson Distribution

For $X \sim \text{Po}(36)$, use the normal approximation with continuity correction to state the calculation for $P(X \leq 40)$.

a. $P(Y < 40.5)$ where $Y \sim N(36, 36)$

b. $P(Y < 40)$ where $Y \sim N(36, 36)$

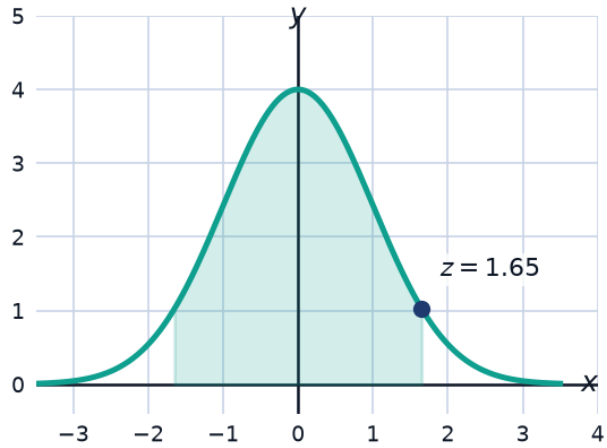
c. $P(Y < 39.5)$ where $Y \sim N(36, 36)$

d. $P(Y < 41)$ where $Y \sim N(36, 6)$

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33 **[Intermediate]** • **Target: Normal Approximation to the Poisson Distribution**

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.



a. 0.980

b. 0.900

c. 0.810

d. 0.990

34 **[Intermediate]** • **Target: Conditions for a Poisson distribution**

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

a. $\frac{19}{27}$

b. $\frac{8}{27}$

c. $\frac{7}{8}$

d. $\frac{2}{3}$

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35 [Intermediate] • Target: Computing Poisson probabilities

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$

b. $16\mathrm{e}^{-4} \approx 0.293$

c. $4\mathrm{e}^{-2} \approx 0.541$

d. $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

36 [Intermediate] • Target: Mean and variance of a Poisson distribution

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. 44

b. 43

c. 11

d. 22

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37 [Intermediate] • Target: Sums of independent Poisson variables

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$; reject H_0 because $-2.0 < -1.645$

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Poisson as an approximation to the binomial • Cross-Lesson: 2.01 Exponentials & Logarithms

$X \sim Po(13)$ and $Y \sim Po(13)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 26

b. 27

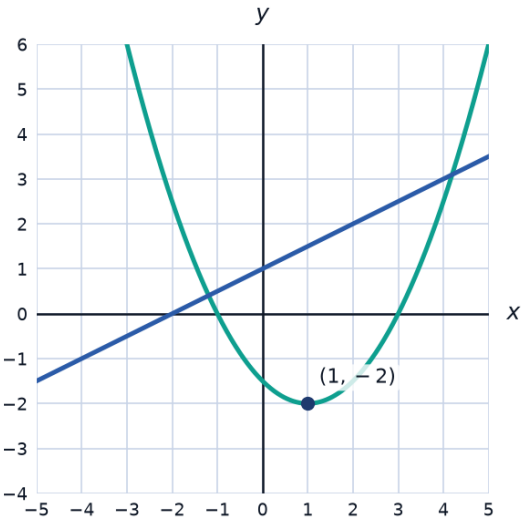
c. 169

d. 0

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39 [Challenging] • Target: Using cumulative Poisson probabilities • Cross-Lesson: 3.01 Advanced Trigonometry

$X \sim Po(15)$ and $Y \sim Po(11)$ are independent. Find the parameter of the distribution of $X + Y$.



a. 27

b. 165

c. 4

d. 26

40 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 4.01 Differentiation

$X \sim Po(18)$ and $Y \sim Po(19)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 342

b. 38

c. 1

d. 37

41 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 5.01 Integration

$X \sim Po(19)$ and $Y \sim Po(25)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 475

b. 45

c. 44

d. 6

42 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 6.01 Sequences & Series

$X \sim Po(18)$ and $Y \sim Po(22)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 396

b. 4

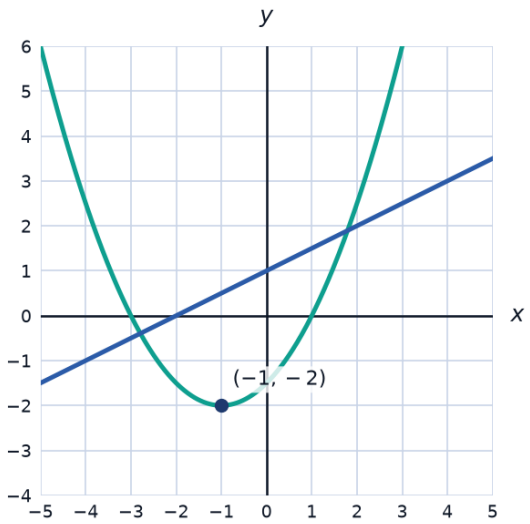
c. 40

d. 41

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43 [Challenging] • Target: Poisson Probability Formula & Conditions • Cross-Lesson: 7.01 Vectors

$X \sim Po(2)$ and $Y \sim Po(14)$ are independent. Find the parameter of the distribution of $X + Y$.



a. 17

b. 16

c. 28

d. 12

44 [Challenging] • Target: Mean, Variance & Linear Combinations • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

$X \sim Po(7)$ and $Y \sim Po(14)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 21

b. 22

c. 7

d. 98

45 [Challenging] • Target: Poisson Approximation to the Binomial • Cross-Lesson: 9.01 Complex Numbers

$X \sim Po(7)$ and $Y \sim Po(15)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 105

b. 22

c. 23

d. 8

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